

A Blockwise Criterion for Ordered Simultaneous Spectral Decompositions in Spectral-Function Critical Cones

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Abstract

This note gives a complete solution of the finite-dimensional blockwise question that appears after Proposition 5.4 of Mohammadi–Sarabi: when do the matrices

$$\Lambda(Y)_{\alpha_m \alpha_m} \quad \text{and} \quad U_{\alpha_m}^\top H U_{\alpha_m}$$

have a simultaneous ordered spectral decomposition? The answer is an explicit uncoupling-and-ordering criterion. Within each eigenspace block of X , group the coordinates by equal eigenvalues of Y . The desired ordered simultaneous decomposition exists exactly when H has no off-diagonal coupling between distinct such Y -eigenspaces and the spectra of the diagonal compressions of H are already ordered from the larger Y -eigenvalue subblocks to the smaller ones. Consequently, Proposition 5.4 yields an explicit matrix-side description of $K_g(X, Y)$ once the vector critical cone $K_\theta(\lambda(X), \lambda(Y))$ is known. Thus the note is a solution of the blockwise simultaneous-decomposition subproblem and partial progress toward a completely closed-form critical-cone formula for arbitrary spectral functions.

Status of the result

The contribution here is deliberately narrow. It does not attempt to compute the vector critical cone $K_\theta(\lambda(X), \lambda(Y))$ for every possible symmetric convex function θ . Instead, it completely characterizes the additional matrix condition in Mohammadi–Sarabi’s critical-cone theorem. In that precise sense it solves the blockwise problem. In the broader sense of a universal, fully expanded description of $K_g(X, Y)$ for all spectral functions $g = \theta \circ \lambda$, it should be read as partial progress: the vector cone remains an input depending on θ .

1 Introduction and notation

Let $\mathbb{S}^n$ denote the space of real $n \times n$ symmetric matrices, equipped with the trace inner product

$$\langle A, B \rangle = \text{tr}(AB).$$

For $X \in \mathbb{S}^n$, let $\lambda(X) = (\lambda_1(X), \dots, \lambda_n(X))$ denote the eigenvalue vector sorted in nonincreasing order. If $I \subset \{1, \dots, n\}$ and $U \in \mathbb{O}^n$, then U_I denotes the submatrix of U formed by the columns indexed by I .

A proper function $g : \mathbb{S}^n \rightarrow \overline{\mathbb{R}}$ is spectral if

$$g(X) = \theta(\lambda(X))$$

for some permutation-invariant function $\theta : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$. For a convex function f and $v \in \partial f(x)$ we use the critical cone convention

$$K_f(x, v) := \{w : df(x)(w) = \langle v, w \rangle\},$$

with the evident Euclidean inner product in the vector case and the trace inner product in the matrix case.

For $X \in \mathbb{S}^n$, define

$$\mathbb{O}^n(X) := \{U \in \mathbb{O}^n : U^\top X U = \text{Diag}(\lambda(X))\}.$$

If $Y \in \partial g(X)$ for a closed convex spectral function $g = \theta \circ \lambda$, then one may choose

$$U \in \mathbb{O}^n(X) \cap \mathbb{O}^n(Y)$$

so that

$$X = U \text{Diag}(\lambda(X)) U^\top, \quad Y = U \text{Diag}(\lambda(Y)) U^\top, \quad \lambda(Y) \in \partial \theta(\lambda(X)).$$

Let

$$\mu_1 > \mu_2 > \cdots > \mu_r$$

be the distinct eigenvalues of X , and set

$$\alpha_m := \{i : \lambda_i(X) = \mu_m\}, \quad m = 1, \dots, r.$$

The first-order directional derivative of the ordered eigenvalue map is then

$$\lambda'(X; H) = (\lambda(U_{\alpha_1}^\top H U_{\alpha_1}), \dots, \lambda(U_{\alpha_r}^\top H U_{\alpha_r})), \quad (1)$$

where the right-hand side denotes concatenation of the listed ordered eigenvalue vectors.

The critical-cone result of Mohammadi–Sarabi [1] may be stated in the following form.

Proposition 1.1 (Mohammadi–Sarabi, Proposition 5.4). *Let $g = \theta \circ \lambda : \mathbb{S}^n \rightarrow \overline{\mathbb{R}}$ be lower semi-continuous and convex, and let $Y \in \partial g(X)$. Fix $U \in \mathbb{O}^n(X) \cap \mathbb{O}^n(Y)$. Then $H \in K_g(X, Y)$ if and only if*

$$\lambda'(X; H) \in K_\theta(\lambda(X), \lambda(Y))$$

and, for every $m = 1, \dots, r$, the two matrices

$$\Lambda(Y)_{\alpha_m \alpha_m} \quad \text{and} \quad U_{\alpha_m}^\top H U_{\alpha_m}, \quad \Lambda(Y) := \text{Diag}(\lambda(Y)), \quad (2)$$

have a simultaneous ordered spectral decomposition.

The purpose of this note is to remove the unresolved phrase in (2). We first solve the corresponding two-matrix problem and then apply it to the blocks in Proposition 1.1.

2 The two-matrix criterion

Definition 2.1 (Ordered simultaneous spectral decomposition). Two matrices $A, B \in \mathbb{S}^k$ have a *simultaneous ordered spectral decomposition* if there exists $Q \in \mathbb{O}^k$ such that

$$Q^\top A Q = \text{Diag}(\lambda(A)), \quad Q^\top B Q = \text{Diag}(\lambda(B)).$$

Thus one common orthonormal basis diagonalizes both matrices and orders both lists of eigenvalues nonincreasingly.

Theorem 2.2 (Pair criterion). *Let $A, B \in \mathbb{S}^k$. Let*

$$\nu_1 > \nu_2 > \cdots > \nu_s$$

be the distinct eigenvalues of A . Write

$$E_j := \ker(A - \nu_j I), \quad P_j := \text{the orthogonal projector onto } E_j.$$

Then A and B have a simultaneous ordered spectral decomposition if and only if both of the following conditions hold:

(a) *B reduces the eigenspaces of A :*

$$P_i B P_j = 0 \quad (i \neq j). \quad (3)$$

(b) *The spectra of the restrictions of B to these eigenspaces are ordered compatibly with the decreasing eigenvalues of A :*

$$\lambda_{\min}(B|_{E_i}) \geq \lambda_{\max}(B|_{E_j}) \quad \text{whenever } i < j. \quad (4)$$

Equivalently, it is enough to require (4) for adjacent pairs $j, j+1$.

Proof. Assume first that A and B have a simultaneous ordered spectral decomposition. Choose $Q \in \mathbb{O}^k$ such that

$$Q^\top A Q = \text{Diag}(\lambda(A)), \quad Q^\top B Q = \text{Diag}(\lambda(B)).$$

For each j , the columns of Q corresponding to the repeated value ν_j in the ordered diagonal $\text{Diag}(\lambda(A))$ form an orthonormal basis of E_j . The same columns are eigenvectors of B , because $Q^\top B Q$ is diagonal. Therefore B maps E_j into E_j for every j , which is equivalent to (3). Moreover, since the diagonal entries of $Q^\top B Q$ are $\lambda_1(B) \geq \cdots \geq \lambda_k(B)$, all eigenvalues of $B|_{E_i}$ occur before all eigenvalues of $B|_{E_j}$ whenever $i < j$. Hence every eigenvalue of $B|_{E_i}$ is at least every eigenvalue of $B|_{E_j}$, proving (4).

Conversely, assume (3) and (4). Since B leaves each E_j invariant and $B|_{E_j}$ is self-adjoint, choose an orthonormal eigenbasis of $B|_{E_j}$ in E_j , ordered so that the corresponding eigenvalues are nonincreasing. Concatenate these bases in the order

$$E_1, E_2, \dots, E_s,$$

and let Q be the resulting orthogonal matrix. Then

$$Q^\top A Q = \text{Diag}(\lambda(A)),$$

because A is scalar with value ν_j on E_j and $\nu_1 > \cdots > \nu_s$. Also $Q^\top B Q$ is diagonal by construction. The diagonal entries of $Q^\top B Q$ are sorted inside each E_j block; condition (4) prevents an inversion between two different E -blocks. Hence the entire diagonal is $\lambda(B)$ in nonincreasing order, and so $Q^\top B Q = \text{Diag}(\lambda(B))$.

Finally, the adjacent-pair formulation is equivalent because the inequalities

$$\lambda_{\min}(B|_{E_j}) \geq \lambda_{\max}(B|_{E_{j+1}}), \quad j = 1, \dots, s-1,$$

chain together to give (4) for all $i < j$. □

Remark 2.3 (Commuting is not enough). For real symmetric matrices, condition (3) is equivalent to $AB = BA$. Thus commutativity gives ordinary simultaneous orthogonal diagonalizability. The extra condition (4) is precisely what makes the diagonalization ordered for both matrices. For example,

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

commute, but they do not have a simultaneous ordered spectral decomposition.

Corollary 2.4 (Fan equality form). *For $A, B \in \mathbb{S}^k$, the following are equivalent:*

- (i) *A and B have a simultaneous ordered spectral decomposition;*
- (ii) *equality holds in Fan's trace inequality,*

$$\langle A, B \rangle = \langle \lambda(A), \lambda(B) \rangle.$$

Consequently, the structural conditions in Theorem 2.2 are exactly the equality conditions for Fan's inequality.

Proof. Fan's trace inequality for real symmetric matrices [3] gives

$$\langle A, B \rangle \leq \langle \lambda(A), \lambda(B) \rangle,$$

and its equality case is precisely simultaneous ordered spectral decomposability; see, for example, Theobald's trace inequality theorem [2]. Combining that equality characterization with Theorem 2.2 gives the final sentence. $\square$

3 The blockwise solution

Return to the setting of Proposition 1.1. For each m , let

$$\eta_{m,1} > \eta_{m,2} > \cdots > \eta_{m,s_m}$$

be the distinct entries of the vector $\lambda(Y)_{\alpha_m}$, and define

$$\gamma_{m,j} := \{i \in \alpha_m : \lambda_i(Y) = \eta_{m,j}\}, \quad j = 1, \dots, s_m. \quad (5)$$

These sets partition α_m . Since $\lambda(Y)$ is sorted nonincreasingly, the sets $\gamma_{m,j}$ appear in the order of decreasing $\eta_{m,j}$.

Theorem 3.1 (Blockwise criterion). *Fix $H \in \mathbb{S}^n$ and $U \in \mathbb{O}^n(X) \cap \mathbb{O}^n(Y)$. For every $m = 1, \dots, r$, the matrices*

$$\Lambda(Y)_{\alpha_m \alpha_m} \quad \text{and} \quad U_{\alpha_m}^\top H U_{\alpha_m}$$

have a simultaneous ordered spectral decomposition if and only if the following two conditions hold for every m :

- (a) *H has no coupling between distinct Y -eigenvalue subblocks inside the same X -eigenvalue block:*

$$U_{\gamma_{m,j}}^\top H U_{\gamma_{m,\ell}} = 0 \quad \text{whenever } j \neq \ell. \quad (6)$$

- (b) *The spectra of the compressed direction blocks are ordered according to the decreasing values of $\lambda(Y)$:*

$$\lambda_{\min}\left(U_{\gamma_{m,j}}^\top HU_{\gamma_{m,j}}\right) \geq \lambda_{\max}\left(U_{\gamma_{m,j+1}}^\top HU_{\gamma_{m,j+1}}\right), \quad j = 1, \dots, s_m - 1. \quad (7)$$

Equivalently, (7) may be required for every pair $j < \ell$ with the right side using $\gamma_{m,\ell}$.

When these conditions hold, a simultaneous ordered diagonalizer on the m th block is obtained by choosing, for each j , an orthogonal matrix $R_{m,j}$ such that

$$R_{m,j}^\top (U_{\gamma_{m,j}}^\top HU_{\gamma_{m,j}}) R_{m,j} = \text{Diag}\left(\lambda(U_{\gamma_{m,j}}^\top HU_{\gamma_{m,j}})\right)$$

and setting

$$Q_m := \text{Diag}(R_{m,1}, \dots, R_{m,s_m}).$$

Proof. Fix m and abbreviate

$$A_m := \Lambda(Y)_{\alpha_m \alpha_m}, \quad B_m := U_{\alpha_m}^\top HU_{\alpha_m}.$$

The matrix A_m is diagonal. Its distinct eigenvalues are exactly

$$\eta_{m,1} > \eta_{m,2} > \dots > \eta_{m,s_m},$$

and the eigenspace corresponding to $\eta_{m,j}$ is the coordinate subspace indexed by $\gamma_{m,j}$. Therefore the projector condition $P_i B_m P_j = 0$ in Theorem 2.2 is precisely

$$(B_m)_{\gamma_{m,i} \gamma_{m,j}} = 0 \quad (i \neq j),$$

which is (6). The ordering condition in Theorem 2.2 becomes exactly (7), because the restriction of B_m to the coordinate subspace $\gamma_{m,j}$ is

$$U_{\gamma_{m,j}}^\top HU_{\gamma_{m,j}}.$$

Applying Theorem 2.2 to (A_m, B_m) for each m proves the equivalence.

The stated construction of Q_m is the constructive half of Theorem 2.2. The zero-coupling condition makes B_m block diagonal with blocks indexed by $\gamma_{m,1}, \dots, \gamma_{m,s_m}$. The matrices $R_{m,j}$ diagonalize these blocks in nonincreasing spectral order, and (7) ensures that concatenating the ordered lists gives the entire vector $\lambda(B_m)$ in nonincreasing order. Since A_m is scalar on each $\gamma_{m,j}$, this block-diagonal Q_m also keeps A_m in the ordered diagonal form $\text{Diag}(\lambda(A_m))$. $\square$

Corollary 3.2 (Critical-cone consequence). *Under the hypotheses of Proposition 1.1, fix $U \in \mathbb{O}^n(X) \cap \mathbb{O}^n(Y)$ and form the sets α_m and $\gamma_{m,j}$ as above. Then $H \in K_g(X, Y)$ if and only if all three conditions hold:*

(i)

$$\lambda'(X; H) \in K_\theta(\lambda(X), \lambda(Y));$$

(ii)

$$U_{\gamma_{m,j}}^\top HU_{\gamma_{m,\ell}} = 0 \quad \text{for every } m \text{ and every } j \neq \ell;$$

(iii)

$$\lambda_{\min}\left(U_{\gamma_{m,j}}^\top HU_{\gamma_{m,j}}\right) \geq \lambda_{\max}\left(U_{\gamma_{m,j+1}}^\top HU_{\gamma_{m,j+1}}\right) \quad \text{for every } m \text{ and } j = 1, \dots, s_m - 1.$$

Thus the only matrix-side condition in Proposition 1.1 is the explicit block-uncoupling and spectral-ordering condition above.

Proof. This is Proposition 1.1 with the simultaneous ordered decomposition requirement replaced by Theorem 3.1. $\square$

4 A coordinate-free formulation

The preceding conditions are independent of the particular simultaneous ordered diagonalizer U . This is clearest in terms of joint spectral projectors.

Let $E_X(\mu_m)$ denote the orthogonal projector onto $\ker(X - \mu_m I)$, and let $E_Y(\eta)$ denote the orthogonal projector onto $\ker(Y - \eta I)$. Since X and Y are simultaneously diagonalizable in the present setting, these projectors commute. Define

$$\Pi_{m,j} := E_X(\mu_m)E_Y(\eta_{m,j}), \quad V_{m,j} := \text{range } \Pi_{m,j}. \quad (8)$$

Then $V_{m,j} = \text{range } U_{\gamma_{m,j}}$.

Corollary 4.1 (Basis-free form). *The blockwise simultaneous ordered decomposition condition in Proposition 1.1 is equivalent to the following coordinate-free conditions, for every m :*

$$(a) \quad \Pi_{m,j} H \Pi_{m,\ell} = 0 \quad (j \neq \ell); \quad (9)$$

(b)

$$\lambda_{\min}((\Pi_{m,j} H \Pi_{m,j})|_{V_{m,j}}) \geq \lambda_{\max}((\Pi_{m,j+1} H \Pi_{m,j+1})|_{V_{m,j+1}}), \quad j = 1, \dots, s_m - 1. \quad (10)$$

Proof. The equality $V_{m,j} = \text{range } U_{\gamma_{m,j}}$ identifies the matrix of the restricted operator $(\Pi_{m,j} H \Pi_{m,j})|_{V_{m,j}}$ in the basis $U_{\gamma_{m,j}}$ with $U_{\gamma_{m,j}}^\top H U_{\gamma_{m,j}}$. Similarly, the operator $\Pi_{m,j} H \Pi_{m,\ell} : V_{m,\ell} \rightarrow V_{m,j}$ has matrix $U_{\gamma_{m,j}}^\top H U_{\gamma_{m,\ell}}$ in the corresponding orthonormal bases. Thus (9) and (10) are exactly (6) and (7). $\square$

Remark 4.2 (Algebraic projectors). The projectors in (8) can be written directly as polynomials in X and Y . For example,

$$E_X(\mu_m) = \prod_{p \neq m} \frac{X - \mu_p I}{\mu_m - \mu_p},$$

and analogously for $E_Y(\eta)$ over the distinct eigenvalues of Y . Hence the criterion can be checked without making any arbitrary choice of basis inside joint eigenspaces.

5 Fan-equality certificate on each block

Corollary 2.4 gives a compact scalar certificate for the same condition. For a fixed m , set again

$$A_m := \Lambda(Y)_{\alpha_m \alpha_m}, \quad B_m := U_{\alpha_m}^\top H U_{\alpha_m}.$$

Then the m th block condition is equivalent to

$$\langle A_m, B_m \rangle = \langle \lambda(A_m), \lambda(B_m) \rangle. \quad (11)$$

Since $A_m = \text{Diag}(\lambda(Y)_{\alpha_m})$, this is

$$\text{tr}\left(\text{Diag}(\lambda(Y)_{\alpha_m}) U_{\alpha_m}^\top H U_{\alpha_m}\right) = \left\langle \lambda(Y)_{\alpha_m}, \lambda(U_{\alpha_m}^\top H U_{\alpha_m}) \right\rangle. \quad (12)$$

The structural content of (12) is exactly (6) plus (7).

6 Special cases and comparison with the semidefinite-cone example

- (a) If $\lambda(Y)$ is constant on α_m , then $s_m = 1$. There are no off-diagonal subblocks and no ordering inequalities. The condition is automatic, as expected, because $\Lambda(Y)_{\alpha_m \alpha_m}$ is a scalar multiple of the identity.
- (b) If the entries of $\lambda(Y)_{\alpha_m}$ are all distinct, then every $\gamma_{m,j}$ is a singleton. The condition says exactly that $U_{\alpha_m}^\top H U_{\alpha_m}$ is diagonal and that its diagonal entries are nonincreasing in the same order as the entries of $\lambda(Y)_{\alpha_m}$.
- (c) Ordinary simultaneous diagonalizability corresponds only to (6). Ordered simultaneous diagonalizability requires the additional inequalities (7).

The example following Proposition 5.4 in Mohammadi–Sarabi is explained immediately by the criterion. There $n = 3$,

$$X = \text{Diag}(1, 0, 0), \quad Y = \text{Diag}(0, 0, -1), \quad U = I.$$

The only nontrivial X -eigenvalue block is $\alpha_2 = \{2, 3\}$, and inside it the Y -eigenvalue subblocks are

$$\gamma_{2,1} = \{2\}, \quad \gamma_{2,2} = \{3\}.$$

Thus the blockwise ordered-decomposition condition forces

$$H_{23} = H_{32} = 0$$

and the scalar ordering condition

$$H_{22} \geq H_{33}.$$

After the vector critical-cone condition for $\theta = \delta_{\mathbb{R}_+^3}$ is also imposed, this recovers the familiar description

$$H \in K_{\mathbb{S}_+^3}(X, Y) \iff H = \begin{pmatrix} * & * & * \\ * & a & 0 \\ * & 0 & 0 \end{pmatrix}, \quad a \geq 0.$$

In particular, the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & -1/2 \\ 0 & -1/2 & 0 \end{pmatrix}$$

fails the critical-cone test for the matrix-side reason $H_{23} \neq 0$, exactly as predicted by (6).

7 Conclusion

The simultaneous ordered spectral decomposition condition in Proposition 1.1 has no remaining hidden content: it is equivalent to the explicit block-zero conditions (6) and the cross-block spectral-ordering inequalities (7). Therefore the blockwise issue raised after Proposition 5.4 is completely resolved.

The resulting critical-cone formula is nevertheless only as explicit as the vector cone $K_\theta(\lambda(X), \lambda(Y))$. For many symmetric functions θ this vector cone is easy to compute, and Corollary 3.2 then gives a full matrix critical cone. For a completely arbitrary convex symmetric θ , the present note should be viewed as partial progress toward a universal closed-form expression, not as a computation of every possible vector critical cone.

References

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